EVALUATION REPORT / PJDSC 2026

BANTAY-BAHA

Flood-linked dengue surge risk and storm-driven loss of hospital access, ranked by province for any typhoon track. Results of the full pipeline run on 501 verified source files.

PJDSC 2026

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WHAT WE BUILT, WHAT THE DATA SAYS

Bantay-Baha joins three graphs into one ranking: a flood-exposure layer (rain, river routing, observed and mapped flood), a lead-lag outbreak graph learned with PCMCI+, and a road-plus-ferry access graph that a typhoon cuts. A quantile forecast of dengue gives the probability **R** that a province's next four weeks exceed its usual bad month; two access methods give the share **A** of people cut off from a hospital; priority is **P = R × A**. Every number below was computed by `build.bat` from the raw files.

0.54
relative WIS of XGBoost vs seasonal naive, 1-4 weeks (target below 1)

0.44
median ensemble with the three benchmarks: the best mean score

85%
XGBoost 90% interval coverage (target 80-95%)

98
pre-computed storm scenarios: 4 typhoons x 15 tracks + 38 more storms

FINDINGS

- **The forecast beats the bar.** Multi-quantile XGBoost reaches relative WIS 0.42 at one week and 0.66 at four over 34,050 rolling-origin forecasts in 2018-2019.
- **Benchmarked, and beaten on the mean.** Zero-shot Chronos-2 and TabPFN-TS and a graph network (MPNN-LSTM) reach relative WIS 0.494-0.504, all below XGBoost; their median ensemble with XGBoost is best (0.444, 90% coverage 90%). On the *median* WIS XGBoost is best (6.23 vs 6.54-6.83): the benchmarks win on high-count epidemic weeks. Storm scenarios still use XGBoost, the only model that takes the storm track.
- **Two honest null results.** Lead-lag graph features do not help (graph / no-graph 0.999), nor do rain, river, storm and flood features (1.003). At 1-4 weeks, recent cases and the season carry the skill.
- **Surge probability R, recalibrated.** Isotonic regression on past rolling-origin forecasts only lowers the Brier score from 0.089 to 0.082 (base rate 0.216) and removes the model's under-confidence in the middle range.
- **The ranking points at the right places.** ρ(R, next-4-week incidence) = 0.68 (Ompong), 0.73 (Tisoy); 7 of Ompong's top-10 provinces surged against a 40% base rate.
- **Access loss, two methods.** For the observed tracks (roads cut by Sentinel-1 observed flood) 0.3 M-1.0 M people lose their route on the road-and-ferry network and 0.6 M-1.6 M on the MAP friction raster; the gap is reported as a range. Gale swaths cancel 42-100 of 108 RoRo routes; Kendall's τ under nine setting changes is at least 0.99.
- **Leptospirosis (stretch).** In 12 cities, rain is followed by leptospirosis 2-5 weeks later (peak IRR 1.40 per +50 mm/week at lag 4); not extrapolated to provinces.

Not in these results: System Usability Scale responses from target users (the test is built into the dashboard); the DOH facility registry (hospitals are the OSM + NOAH fallback). Scorecard: page 17.

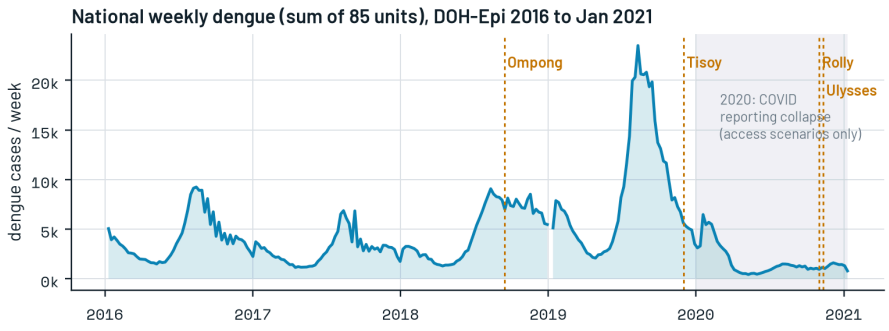


Figure 1. The forecast target: DOH-Epi weekly dengue summed over the 85 harmonised units, with the four backtest typhoons. 2020 is used for access scenarios only.

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DATA AND ACQUISITION

501 files from 22 sources (7.0 GB) are listed in `data/raw/MANIFEST.csv` with size and sha256; all were verified on disk before this build. Open sources come from `scripts/download_data.py`, Earth Engine sources from `scripts/download_gee.py`, and the team shares them through a private bucket (`setup.bat`, every file checked against the manifest). The two manual sources were not needed (last rows).

FOLDER	DATASET	ROLE	FROM	FILES	MB
dengue_hdx	DOH-Epi dengue, province/city x week 2016–Jan 2021	forecast target	HDX	1	2
census2020	2020 CPH population by municipality	population	HDX	1	1
cod_ab	COD-AB boundaries ADM0-4 (v03, 2025)	join keys, geometry	HDX	2	370
ibtracs	IBTrACS v04r01 Western Pacific	storm exposure, swaths	NOAA NCEI	1	115
chirps	CHIRPS v2.0 daily 0.05°, 2016–2020	rain features, routing	IRI Data Library	5	563
imerg	GPM IMERG V07 daily 0.1°	rain features	Earth Engine	60	68
hydrorivers	HydroRIVERS v1.0 Australasia	river routing	HydroSHEDS	2	139
noah	UP NOAH flood hazard 5/25-yr, 139 provinces	static susceptibility, road proxy	UP NOAH (Drive)	142	4,733
copernicus_ems	EMSR312 / EMSR478 vector packages	observed flood (Ompong AOIs)	Copernicus EMS	16	28
worldpop	WorldPop 2020 100 m UN-adjusted	population grid	WorldPop	1	159
meta_hrsl	Meta HRSL under-5 / 60+	vulnerable groups cut off	HDX	2	112
osm	OpenStreetMap Philippines PBF	road network	Geofabrik	1	611
roroph	roroph RoRo routes (108)	ferry edges (2024–26 proxy)	pinasr/roroph	2	<1
hotosm_health	HOT OSM health facilities	hospital nodes (fallback)	HDX	1	2
noah_facilities	UP NOAH critical facilities	hospital nodes (fallback)	UP NOAH	1	4
map_friction	MAP friction surface 2019	raster access method	Earth Engine	2	2
gfd	Global Flood Database events	context only (ends 2017)	Earth Engine	1	<1
sentinel1	Sentinel-1 scene inventory	pass dates per storm	Earth Engine	1	<1
s1_flood	Sentinel-1 new-flood masks per typhoon	observed flood	Earth Engine	251	2
cchain	Project CCHAIN PIDSR city-week	leptospirosis, 12 cities	HDX	4	7
mendeley_parreno	Parreño national weekly notifiable diseases	leptospirosis, national	Mendeley Data	1	<1
opendengue	OpenDengue V1.3	national cross-check	OpenDengue	3	110
psgc	*PSA PSGC 2Q 2026*	*not needed: COD-AB pcodes resolve every unit*	*manual*	–	–
nhfr	*DOH NHFR facility registry*	*not used: OSM + NOAH fallback*	*manual*	–	–

STEP 0: ONE UNIT-WEEK TABLE

HARMONISED UNITS

The DOH file has **126 reporting units, not 81 provinces**: provinces mixed with highly urbanised and component cities, plus misspellings (NUEVA VISCAYA, SISQUIJOR, BASILIG CITY = Bislig). COD-AB uses current boundaries (Maguindanao split in 2022, Special Geographic Area in 2024). The reviewed crosswalk in `data/crosswalks/` folds cities into their province and undoes the post-2020 changes, giving **85 units**; every dengue unit resolves and every unit has data. A unit-week counts only if every member reported (NaN otherwise, never 0). Deaths are not used (2016 case fatality 3.9%, far above other years), and 2020 fell to 91k cases against 442k in 2019 under COVID-19, so 2020 is never a forecast target.

POPULATION

The 2020 census table's municipal codes are old 9-digit PSGC and do not join to COD-AB for the highly urbanised cities, post-split Maguindanao and the Special Geographic Area. Its own *Province* column already places cities in their province, so units are summed on it: 108.67 M people.

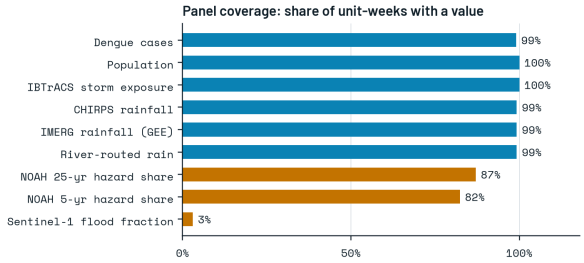


Figure 2. Coverage of the 85 x 262 panel. Sentinel-1 is filled only for the four landfall windows, by design.

LAYER 1 INPUTS

Rain: CHIRPS and IMERG agree (weekly r = 0.87). River routing accumulates CHIRPS rain down the HydroRIVERS network. NOAH 25-yr hazard covers 74 of 85 units; the median unit has 20% of its people in a hazard zone. Units with no map are **missing, not zero**: Basilan, Batanes, Biliran, Camiguin, Davao Occidental, Guimaras, Northern Samar, Romblon, Siquijor, Sulu, Tawi-Tawi.

Storm exposure: every Western Pacific track 2016-2020 from IBTrACS, gale radius from USA R34, else JMA R30, else a fixed 150 km. USA R34 is missing on 29% of Rolly's gale-strength fixes (JMA fills them); the fixed fallback never triggers for the four storms.

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METHODS AND EVALUATION PROTOCOL

LAYER 2: LEAD-LAG OUTBREAK GRAPH

log(1 + incidence), 52-week seasonal difference, standardised inside the training window. PCMCI+ (Runge, 2020) with partial-correlation tests, lags 1-3 weeks, pc_alpha 0.01; lagged inter-province edges are kept when the Benjamini-Hochberg (1995) q-value is below 0.05 (on pure-noise checks PCMCI+ without this step returned many false edges). PageRank (Page et al., 1999) on the reversed graph scores sentinels; Louvain (Blondel et al., 2008) finds communities; lagged cross-correlation with the same FDR is the baseline graph.

FORECAST AND SURGE PROBABILITY

One pooled XGBoost (Chen & Guestrin, 2016) per horizon (1-4 weeks), multi-quantile objective at the 23 CDC FluSight levels, on log(1 + incidence), back-transformed to cases. Features: lagged incidence, same week last year, week of year, CHIRPS / IMERG / river rain lags, storm exposure, NOAH hazard, Sentinel-1 flood and leader signals from the graph; SHAP (Lundberg & Lee, 2017) gives the "drivers". A direct 4-week model gives **R**, the probability of exceeding the unit's 90th percentile of 4-week incidence in the training window. R is then **recalibrated** by isotonic regression (Zadrozny & Elkan, 2002) fitted only on earlier rolling-origin forecasts whose outcome was known at the origin.

Benchmarks, scored on exactly the same rows: Chronos-2 (Ansari et al., 2025) and TabPFN-TS (Hoo et al., 2025) zero-shot, each origin's forecast from that unit's own past only; an MPNN-LSTM graph network (Panagopoulos et al., 2021) trained on the same 13-week rolling refits; and the element-wise median of the 23 quantiles of XGBoost and the three (the ensemble).

EVALUATION

Rolling origin, never a random split. From January 2018 to December 2019 the model, graph and thresholds are refit every 13 weeks on data up to the refit date, then forecast every origin until the next refit. Scores: weighted interval score on case counts (Bracher et al., 2021), relative WIS against a seasonal-naive forecast, 50% and 90% coverage, and an online conformal 90% band from past errors only (Gibbs & Candès, 2021).

LAYER 3: ACCESS, TWO METHODS

Network. OSM highways become a graph of junctions; RoRo routes are ferry edges (crossing time plus half the headway); hospitals are HOT OSM plus NOAH name-matched facilities (2,199). WorldPop (Tatem, 2017) is summed to 1 km cells, each joined to the nearest road node; a multi-source Dijkstra gives hours to the nearest hospital. **Raster.** Cost-distance from the same hospitals over the MAP motorised friction surface (Weiss et al., 2020), sea impassable except along the same ferry lines. **Storms** apply the same rules to both: ferries crossing the gale swath are cancelled; roads (or cells) in observed flood (Sentinel-1, observed tracks) or in NOAH high hazard inside the swath (shifted tracks) are cut, medium hazard slows them. A cell is cut off if it loses its route, or its trip at least doubles and grows by over an hour (after Petricola et al., 2022). The network gives A, P and the tier; the per-unit minimum and maximum of the two methods give A_lo-A_hi and P_lo-P_hi. Under-5 and 60+ people come from Meta HRSL (Tiecke et al., 2017) summed to the same cells; a cell's vulnerable people are cut off when the cell is (network rule).

FORECAST SKILL (XGBOOST)

HORIZON	N	WIS XGB	WIS NO-GRAPH	WIS NAIVE	REL WIS	GRAPH / NO-GRAPH	COV 50	COV 90	COV 90 CONFORMAL
1 wk	8,640	14.7	14.7	35.1	**0.42**	0.998	0.47	0.89	0.88
2 wk	8,555	18.1	18.0	35.4	**0.51**	1.002	0.44	0.86	0.87
3 wk	8,470	21.0	21.1	35.6	**0.59**	0.998	0.42	0.84	0.86
4 wk	8,385	23.6	23.6	35.9	**0.66**	0.999	0.40	0.82	0.85
all	**34,050**	**19.3**	**19.3**	**35.5**	**0.54**	**0.999**	**0.43**	**0.85**	**0.87**

Targets met. Relative WIS is below 1 at every horizon (0.42 to 0.66) and 90% coverage is inside 80-95% at every horizon (four weeks is close to the floor at 0.82). The 50% intervals are a little narrow (0.40-0.47). The online conformal band reaches similar coverage, a sign the quantiles are not badly miscalibrated.

WHAT CARRIES THE SKILL

COMPARISON	RELATIVE WIS	READING
XGBoost vs seasonal naive	0.544	46% lower WIS than "same week last year"
with vs without graph leader features	0.999	no gain: the graph has 0 links in the 2018 windows and 2-6 in 2019
with vs without rain / river / storm / flood	1.003	0.544 vs 0.542 against seasonal naive, same rows

With so few significant links, leader signals are empty for most provinces. The rain result has two likely reasons, not tested: rain moves with the season in a four-year panel, so the season features already carry it, and any effect at lags beyond four weeks is outside this horizon.

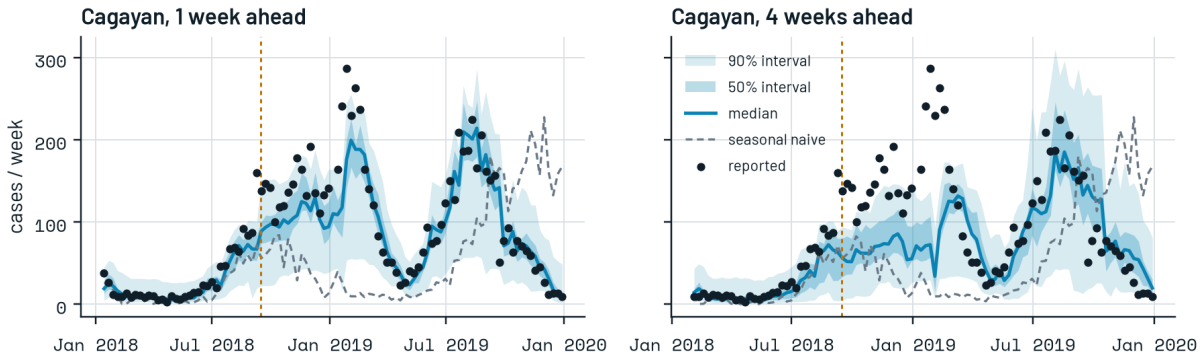


Figure 3. Rolling-origin forecasts for Cagayan (Ompong's landfall province, dashed line), one and four weeks ahead.

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MODEL COMPARISON

Every model scored on the same 34,050 rolling-origin rows (2018-2019, horizons 1-4, WIS on case counts). Relative WIS is against the seasonal naive (below 1 is better) and against XGBoost.

MODEL	KIND	MEAN WIS	MEDIAN WIS	REL 1 WK	REL 4 WK	REL WIS	VS XGB	COV 50	COV 90
median ensemble	ensemble	**15.7**	**6.29**	**0.331**	**0.555**	**0.444**	**0.816**	**0.48**	**0.90**
Chronos-2, zero-shot	benchmark	17.5	6.54	0.376	0.612	0.494	0.908	0.43	0.88
MPNN-LSTM, graph neural net	benchmark	17.6	6.83	0.384	0.604	0.495	0.910	0.48	0.88
TabPFN-TS, zero-shot	benchmark	17.9	6.72	0.349	0.659	0.504	0.926	0.46	0.87
XGBoost, no rain/river/storm/flood	ablation	19.2	6.26	0.420	0.656	0.542	0.997	0.43	0.85
XGBoost + graph (ours)	model	**19.3**	**6.23**	**0.418**	**0.658**	**0.544**	**1.000**	**0.43**	**0.85**
XGBoost, no graph	ablation	19.3	6.24	0.418	0.658	0.544	1.001	0.43	0.85
seasonal naive	baseline	35.5	14.97	1.000	1.000	1.000	1.839	0.42	0.85

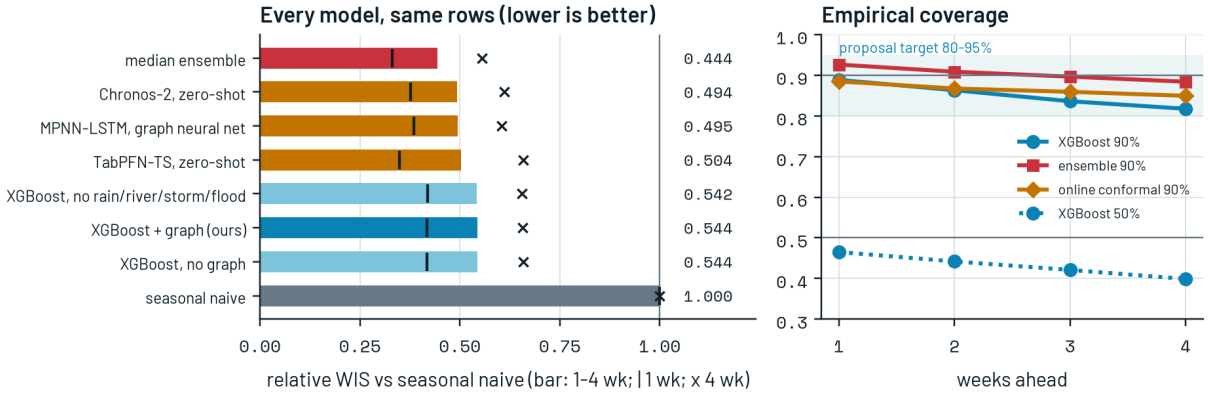


Figure 4. Left: relative WIS of every model over 1-4 weeks (ticks: 1 and 4 weeks). Right: coverage against the nominal lines and the proposal's 80-95% band.

READING THE COMPARISON

- **Mean vs median.** Every benchmark beats XGBoost on mean WIS, and the ensemble is best. On the median row XGBoost is best (6.23 cases vs 6.54-6.83; ensemble 6.29). Split at the 90th percentile of weekly cases (198): below it the benchmarks' mean WIS is 1.02-1.08 times XGBoost's, above it 0.80-0.81 times. The better means come from the large-count epidemic weeks; the ensemble (0.94 / 0.71) gains in both.
- **Chronos-2** as first specified, on log(1 + incidence), collapsed on one-week reporting spikes (relative WIS 112). The headline run uses the incidence scale with past-only covariates; it was chosen *after* seeing that failure. Without covariates it scores 0.480, so the covariates do not help.
- **TabPFN-TS** runs univariate (it ignores past-only covariates) with the ungated TabPFN v2 regressor; the default v3.5 weights need a login and licence and were not used. A lagged-covariate variant scores 0.555.
- **MPNN-LSTM** uses border adjacency (133 edges) plus RoRo routes (36 more, a 2024-26 proxy); with no graph it scores 0.492, the same within seed noise, in line with the sparse lead-lag graph.
- **Where it is used.** Benchmarks and the ensemble exist only at the rolling-origin weeks (2018-2019). Storm scenarios, shifted tracks, 2020 and the catalogue use XGBoost with calibrated R, because the zero-shot models take no storm-track input.

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SURGE PROBABILITY R: RECALIBRATED

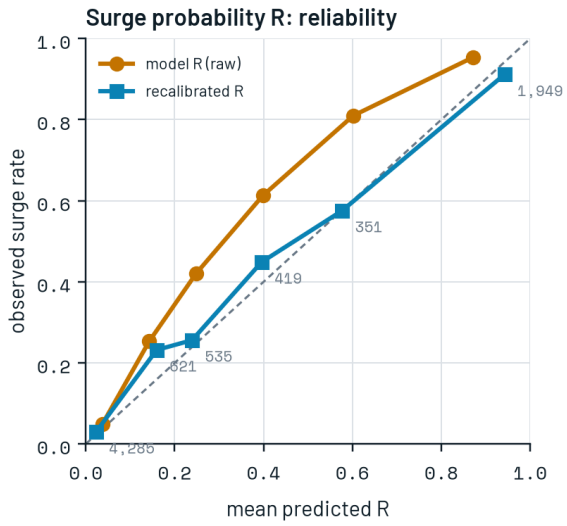


Figure 5. Reliability of R over the 2018-2019 unit-weeks, before (amber) and after (teal) isotonic recalibration; labels: unit-weeks per bin.

The model's R was **under-confident in the middle**: in the bins between 0.1 and 0.7, outcomes came true up to 21% more often than forecast. Isotonic regression maps raw R to observed surge frequency. It is fitted at each origin on earlier forecasts whose 4-week outcome was already known (target week no later than the origin), so it never sees the future. It needs 850 past outcomes first: the 13 origins before 2018-04-08 keep raw R (1,105 rows).

After recalibration the largest middle-bin gap is 7%.

ROWS	N	BASE RATE	MEAN R RAW	MEAN R CAL	BRIER RAW	BRIER CAL	BRIER BASE RATE
all 2018-2019 origins	8,160	0.315	0.254	0.310	0.089	**0.082**	0.216
origins with a calibrator	7,055	0.359	0.286	0.351	0.100	**0.092**	0.230

The base rate is 32%, not 10%, because 2019 was an epidemic year and the thresholds come from earlier, quieter years. The scenario files keep both: R (calibrated, used for P and the tiers) and `R_raw`. Recalibration raises mid-range R, so more provinces reach the provisional tiers (pre-position at R ≥ 0.3, deploy at R ≥ 0.5 with A ≥ 10%): for Ompong mean R rises from 0.22 to 0.29. The tier thresholds still need review with users.

Ties. Isotonic regression is a step function, so provinces share values: for Ompong the 85 units take 18 distinct R values and 4 share the top value (0.99). A ranking by R alone has ties; P = R × A breaks them by access, and the backtest's Spearman correlations use average ranks.

09

THE LEAD–LAG GRAPH



Final window: 84 series, T = 156 weeks after seasonal differencing (Davao Occidental sits out with more than 10% missing weeks). PCMCI+ keeps **8** lagged inter-province links at q < 0.05. Plain lagged cross-correlation with the same FDR keeps **4,265**, because the 2019 national epidemic makes almost every pair co-move.

The proposal's sentinel idea is therefore **weakly supported at province-week level**. The links are statistical dependencies, not evidence of transmission, and several join distant provinces. Likely reasons: weekly province totals average out local waves, four years is short for 84 series, and reporting artefacts dominate small provinces. Synchrony studies such as van Panhuis et al. (2015) worked at longer time scales. The MPNN-LSTM benchmark points the same way: its result does not depend on its graph.

Figure 6. The 8 PCMCI+ links learned on 2016–2019 (teal = leader).

LEADER	FOLLOWER	LAG (WK)	PARTIAL CORR	Q	BOOTSTRAP FREQ
Cavite	Sorsogon	1	0.38	0.001	0.70
Metropolitan Manila Second District	Bulacan	3	0.36	0.006	0.70
Quirino	Surigao del Sur	1	0.36	0.007	0.65
Ilocos Norte	Romblon	1	0.33	0.020	0.75
Metropolitan Manila Third District	Zamboanga del Norte	1	0.33	0.029	0.85
Sulu	Ilocos Sur	1	0.32	0.037	0.55
Camarines Sur	Pangasinan	1	0.31	0.042	0.65
Samar (Western Samar)	Antique	1	0.31	0.045	0.55

OBSERVED FLOOD: SENTINEL-1

Radar sees through typhoon clouds. For each typhoon, new open water is mapped on Google Earth Engine (Gorelick et al., 2017) from Sentinel-1 VH backscatter in 100 m tiles near the track: before = median of days -30 to -3, after = darkest image of days 0 to +10, **both from the same relative orbit**. A pixel is new flood when VH drops by at least 3 dB and ends at most -18 dB; permanent water (JRC), slopes over 5° and specks under 8 pixels are removed (after UN-SPIDER, n.d.). The first version, UN-SPIDER's after/before ratio on dB values with mixed orbits, recovered almost none of the EMS flood and was replaced.

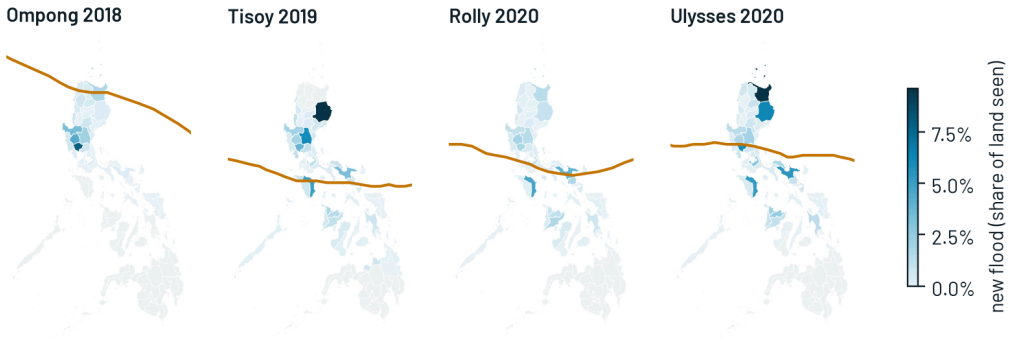


Figure 7. New-flood share of the land Sentinel-1 saw, per unit, with each track.

TYPHOON	PEOPLE IN NEW FLOOD	UNITS >1% FLOODED	MAX UNIT SHARE	MOST PEOPLE FLOODED
Ompong	0.5 M	5	8.0%	Pampanga
Tisoy	1.1 M	14	9.6%	Isabela
Rolly	0.6 M	10	4.7%	Camarines Sur
Ulysses	1.4 M	14	10.2%	Cagayan

PER MUNICIPALITY

TYPHOON	SEEN >20%	>1% FLOODED	MOST PEOPLE IN NEW FLOOD: MUNICIPALITY (PROVINCE), PEOPLE, SHARE	ρ(A, FLOOD)
Ompong	628	131	Candaba (Pampanga) 42k, 34%; Apalit (Pampanga) 34k, 28%; Arayat (Pampanga) 23k, 18%	0.63
Tisoy	1,074	216	Jaen (Nueva Ecija) 37k, 43%; San Mateo (Isabela) 34k, 47%; Alicia (Isabela) 31k, 62%	0.64
Rolly	1,041	173	Naujan (Oriental Mindoro) 23k, 15%; Nabua (Camarines Sur) 22k, 28%; Bula (Camarines Sur) 21k, 18%	0.54
Ulysses	928	249	Tuguegarao City (Cagayan) 45k, 36%; Solana (Cagayan) 44k, 43%; Apalit (Pampanga) 31k, 28%	0.69

ρ: Spearman correlation, over the municipalities the radar saw, between network access loss A and the new-flood share. Access loss follows flood but not one-to-one: a municipality can be flooded and keep a dry road to a hospital.

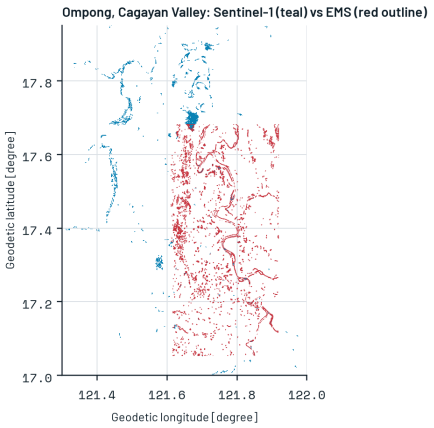


Figure 8. Ompong around Tuguegarao: Sentinel-1 new flood (teal) and Copernicus EMS observed flood (red outlines).

CHECKED AGAINST COPERNICUS EMS

Inside the EMS areas of interest, EMS mapped 108 km² of observed flood and Sentinel-1 flagged 46 km²; 26 km² overlap: **55%** of Sentinel-1 flood there is confirmed by EMS, and Sentinel-1 recovers **24%** of the EMS area. Low recall is expected: EMS used other imagery and dates, while one Sentinel-1 orbit passed here only twice in the window.

ACCESS LOSS UNDER FOUR TYPHOONS

Baseline. 109.4 M people in 1 km cells; on the network 97.3% have a road or ferry route to one of 2,199 hospitals, with a population-weighted mean trip of 0.24 h. The raster reaches 99.5% with a mean of 0.19 h. Results are also written for 1,641 municipalities.

TYPHOON	CUT OFF NETWORK	RASTER	RANGE (UNIT MIN-MAX)	P A UNITS	T P	TOP-10 SHARED	UNDER-5 CUT OFF	60+ CUT OFF
Ompong	0.3 M	0.6 M	0.3-0.6 M	0.80	0.74	7/10	27k	18k
Tisoy	0.7 M	1.3 M	0.7-1.3 M	0.86	0.72	6/10	65k	41k
Rolly	0.4 M	0.7 M	0.3-0.8 M	0.78	0.70	7/10	34k	22k
Ulysses	1.0 M	1.6 M	1.0-1.7 M	0.85	0.73	7/10	96k	63k

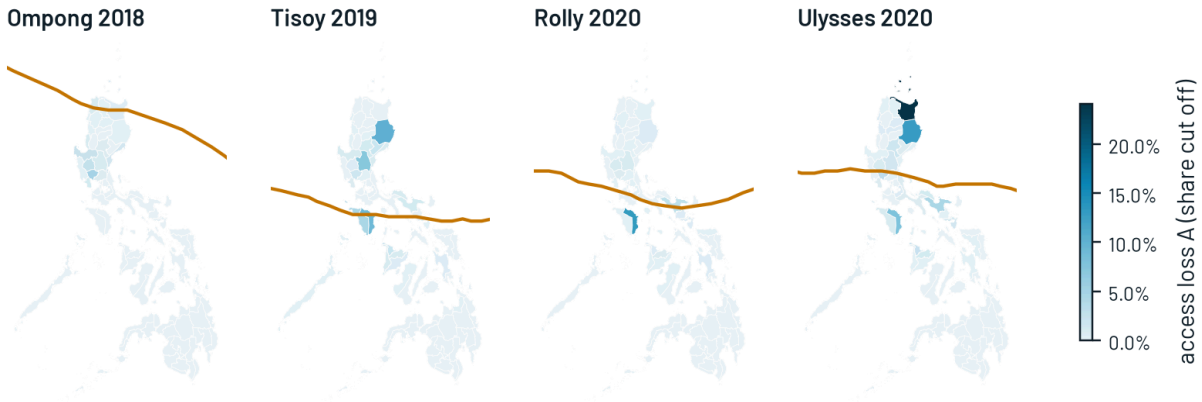


Figure 9. Network access loss A per unit for the observed tracks. Roads are cut by Sentinel-1 observed flood; ferries crossing the gale swath are cancelled.

Observed tracks: roads are cut only where Sentinel-1 saw new water. Radar misses floods under dense vegetation, in built-up areas and between passes, so these are **lower bounds**. The 56 shifted tracks have no image and use the NOAH 25-yr proxy (every high-hazard road in the swath cut): 2.3 M-11.3 M people cut off on the network, an **upper bound**. Ferry routes are 2024-2026 operations, a proxy for 2018-2020. A uses radar that exists about ten days after the forecast origin: it is a retrospective access estimate.

NETWORK VS FRICTION RASTER

The proposal runs a raster method next to the network because OSM completeness is uneven, and reports their disagreement as uncertainty. Both use the same hospitals, population cells, ferry lines and storm rules.

ON A NORMAL DAY

The raster reaches more people (99.5% vs 97.3%) and is faster (population-weighted mean 0.19 h vs 0.24 h; median cell ratio raster / network 0.62): it can cross land without a mapped road. The two still rank places alike: Spearman 0.89 over 321,701 cells that both reach, 0.95 over the 85 units' mean trip.

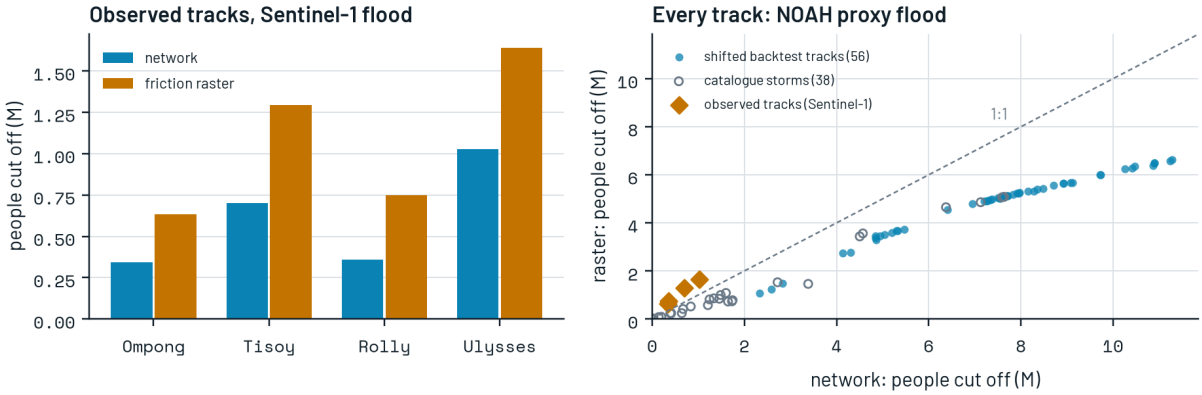


Figure 10. People cut off by each method. Left: observed tracks with Sentinel-1 flood. Right: every track that uses the NOAH proxy (shifted backtest tracks and catalogue storms), with the 1:1 line.

UNDER A STORM

The gap changes sign with the flood source. With Sentinel-1 the raster cuts off more people (0.6 M-1.6 M vs 0.3 M-1.0 M); with the NOAH proxy it cuts off fewer (median ratio 0.66). A likely reason: at 1 km a flooded cell centre closes the whole cell, while radar flood is patchy and the network closes only the flooded road edges; the proxy closes every high-hazard road, which the raster's cell-centre test samples more sparsely. The unit rankings agree well under the proxy (ρ of A 0.91-0.99) and less with radar (0.78-0.86), where 6-7 of the top-10 provinces by P are shared. The dashboard shows A and P with this range. Sensitivity on the raster: Kendall's $\tau \geq 0.999$ for the same nine variants.

VULNERABLE GROUPS

Meta HRSL puts 11.1% of people under five and 6.7% aged 60+; the 2020 census has 10.2% and 8.5%. HRSL therefore **under-counts older people** (and slightly over-counts young children), so the 60+ cut-off numbers are low estimates. Among those cut off, the under-5 share is 7.9%-9.5% and the 60+ share 5.2%-6.2%.

PRIORITY: OMPONG 2018

TYPHOON	FERRY ROUTES CUT	PEOPLE CUT OFF	MEAN A	MEAN R	DEPLOY / PRE-POSITION	TOP PRIORITY
Ompong Mangkhut 2018	42	0.3 M	0.2%	0.29	0 / 22	Pampanga
Tisoy Kammuri 2019	100	0.7 M	0.6%	0.12	0 / 12	Camarines Norte
Rolly Goni 2020	42	0.4 M	0.3%	0.02	0 / 0	Oriental Mindoro
Ulysses Vamco 2020	54	1.0 M	0.8%	0.04	0 / 2	Cagayan

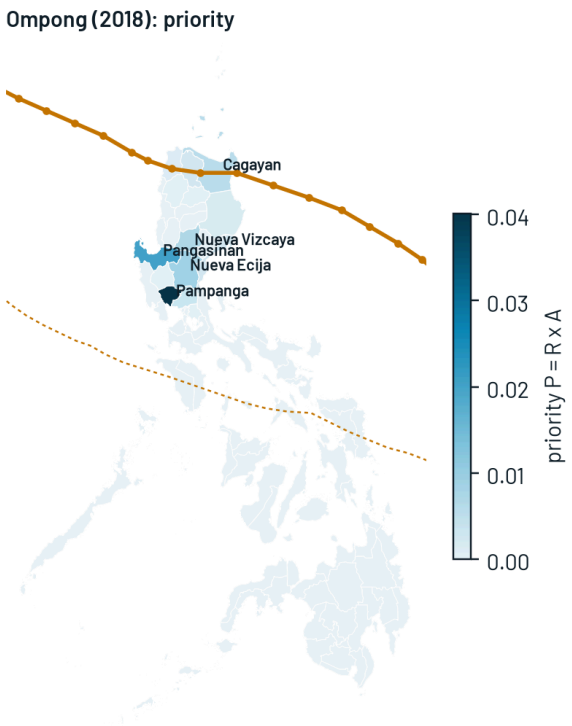


Figure 11. P = R x A for Ompong with its track and gale swath; top 5 labelled.

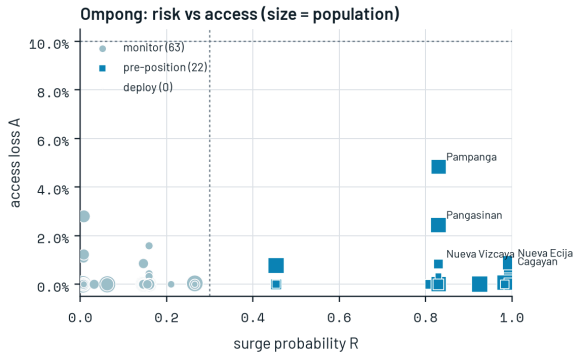


Figure 12. Risk vs access. Dotted lines are the provisional tier thresholds (R = 0.3, A = 10%); marker shape encodes the action tier.

Ompong made landfall in Cagayan on 15 September 2018. The model, trained on data to the landfall week only, puts **Pampanga** first, with 0 provinces in the "deploy mobile clinics" tier and 22 in "pre-position supplies". "Deploy" needs at least 10% of people cut off, which observed flood alone did not produce for any of the four storms; the hazard-proxy shifted tracks of Ompong put 6-7 provinces there.

The dashboard repeats this for every storm and 14 shifted variants of each track (±0.5° and ±1° north or south, ±1° east or west), so a health officer can ask "what if it comes in further south?" and get an answer in seconds.

EVERY STORM, 2017–2020

Beyond the four backtests, the build scores the observed track of every other storm whose gale swath touched Philippine land: **38 storms**. Each uses the latest model fitted before its week (never later data), R calibrated where enough past outcomes exist (24 of 38), and **the NOAH 25-yr proxy for flood** (no radar map exists for them), so their cut-off counts are upper bounds. 9 storms of 2016 are skipped: their model would have under 52 weeks of history.

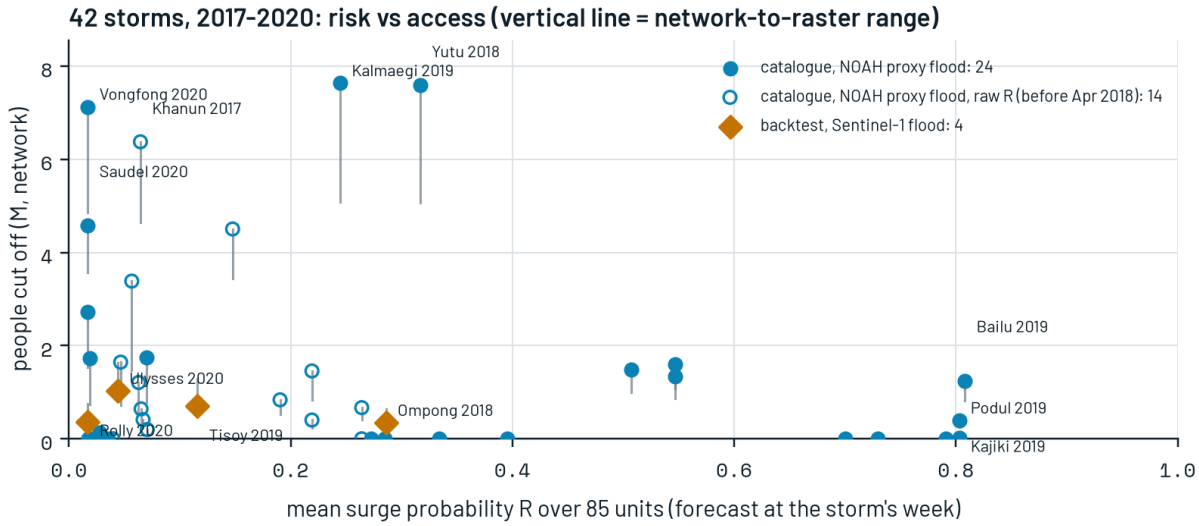


Figure 13. Mean surge probability and people cut off per storm. Diamonds: the four backtests (Sentinel-1 flood, lower bounds); circles: catalogue storms (NOAH proxy, upper bounds); hollow: raw R.

STORM	R	CUT M	RASTER	R×CUT	FERRY	D/P
Nesat 2017	0.26†	0.00	0.00	0k	2	0/27
Haitang 2017	0.26†	0.67	0.42	174k	2	0/28
Hato 2017	0.22†	0.40	0.23	114k	2	0/21
Pakhar 2017	0.22†	1.46	0.84	419k	2	2/21
Mawar 2017	0.19†	0.84	0.52	368k	2	1/19
Doksuri 2017	0.15†	4.50	3.44	1,883k	22	4/9
Khanun 2017	0.06†	6.37	4.65	799k	24	1/3
Damrey 2017	0.06†	1.21	0.58	24k	64	0/4
Haikui 2017	0.07†	0.20	0.09	6k	0	0/6
Kirogi 2017	0.07†	0.41	0.25	10k	12	0/5
Kai-Tak 2017	0.07†	0.64	0.24	23k	30	0/4
Tembin 2017	0.06†	3.38	1.46	52k	38	0/3
Bolaven 2018	0.04†	0.00	0.00	0k	0	0/0
Sanba 2018	0.05†	1.65	0.73	34k	16	0/2
Gaemi 2018	0.27	0.00	0.00	0k	0	0/29
Son-Tinh 2018	0.55	1.60	1.08	1,452k	2	7/43
Unnamed 2018	0.55	1.34	0.86	1,194k	2	7/43
Barijat 2018	0.40	0.00	0.00	0k	2	0/41
Ompong 2018 ◆	0.29	0.34	0.63	223k	42	0/22
Yutu 2018	0.32	7.59	5.06	2,890k	46	8/18
Usagi 2018	0.33	0.00	0.00	0k	0	0/29

STORM	R	CUT M	RASTER	R×CUT	FERRY	D/P
Danas 2019	0.73	0.01	0.00	10k	2	0/66
Lekima 2019	0.79	0.00	0.00	0k	0	0/72
Bailu 2019	0.81	1.24	0.82	999k	2	6/70
Podul 2019	0.80	0.39	0.24	274k	0	1/73
Kajiki 2019	0.80	0.02	0.01	15k	0	0/74
Tapah 2019	0.70	0.00	0.00	0k	2	0/68
Mitag 2019	0.51	1.49	1.00	266k	2	1/52
Nakri 2019	0.28	0.00	0.00	0k	0	0/31
Kalmaegi 2019	0.24	7.63	5.08	1,957k	46	3/27
Tisoy 2019 ◆	0.12	0.70	1.29	28k	100	0/12
Phanfone 2019	0.07	1.74	0.78	131k	76	1/5
Vongfong 2020	0.02	7.13	4.86	62k	58	0/0
Mekkhala 2020	0.04	0.00	0.00	0k	0	0/2
Bavi 2020	0.03	0.14	0.08	1k	2	0/1
Noul 2020	0.02	0.00	0.02	0k	2	0/0
Linfa 2020	0.02	1.73	0.74	30k	46	0/0
Saudel 2020	0.02	4.57	3.56	49k	2	0/0
Molave 2020	0.02	2.72	1.53	37k	46	0/0
Rolly 2020 ◆	0.02	0.36	0.75	5k	42	0/0
Atsani 2020	0.02	0.00	0.00	0k	2	0/0
Ulysses 2020 ◆	0.04	1.03	1.64	10k	54	0/2

R: mean over 85 units; cut: people cut off (M), network and raster; R×cut: expected people at surge risk who are cut off (Σ R × cut off over units); D/P: provinces at deploy / pre-position. ◆ backtest (Sentinel-1); † raw R (before calibration was possible). The largest cut-offs (Kalmaegi 2019, Yutu 2018 and Vongfong 2020) pair with low R; high-R storms (Bailu 2019, Podul 2019 and Kajiki 2019) came in the 2019 epidemic season. Network-raster agreement across units: ρ median 0.96 (min 0.82) over 27 storms with any cut-off.

BACKTEST, SENSITIVITY, BENCHMARK

DID HIGH-PRIORITY PROVINCES SEE MORE CASES?

TYPHOON	ORIGIN	UNITS	SURGED	P(R, OBS)	P(P, OBS)	TOP-10 SURGED	BRIER R	NOTE
Ompong	2018-09-16	85	40%	0.68	0.21	70%	0.161	
Tisoy	2019-12-08	85	5%	0.73	-0.03	0%	0.055	
Rolly	2020-11-01	61	0%	0.41	0.04	0%	0.001	2020: COVID-depressed reporting, indicative only
Ulysses	2020-11-15	85	1%	0.34	0.13	0%	0.021	2020: COVID-depressed reporting, indicative only

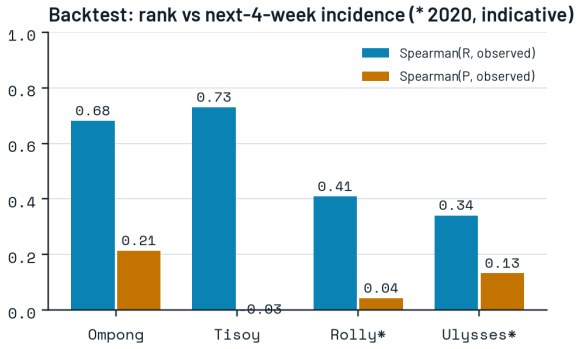


Figure 14. Spearman correlation of each ranking (calibrated R) with observed incidence over the next four weeks, relative to each unit's threshold.

R ranks provinces well for both 2018-2019 storms. P correlates less with cases, as it should: A measures access, not disease. After Ompong, 7 of the 10 highest-priority provinces surged against a 40% base rate. After Tisoy almost nothing surged (the 2019 season was over), so the top-10 rate is 0 and the Brier score is low for the right reason.

SENSITIVITY: KENDALL'S τ OF P AGAINST THE BASE SETTINGS (NETWORK)

VARIANT	OMPONG τ	TISOY τ	ROLLY τ	ULYSSES τ	MEAN A (OMPONG)
cut-off ratio 1.5	1.000	1.000	1.000	1.000	0.23%
cut-off ratio 3	1.000	0.996	0.997	1.000	0.23%
ferry rule: port in swath	1.000	1.000	1.000	1.000	0.23%
gale fallback 100 km	1.000	1.000	1.000	1.000	0.23%
gale fallback 200 km	1.000	1.000	1.000	1.000	0.23%
medium hazard slow x0.25	1.000	1.000	1.000	1.000	0.23%
medium hazard slow x0.75	1.000	1.000	1.000	1.000	0.23%
speed x0.8	1.000	1.000	0.993	0.998	0.23%
speed x1.2	1.000	0.995	0.989	0.999	0.23%

Every variant changes A a little but none reorders the ranking: P's spread is dominated by R. The same variants on the raster method give $\tau \geq 0.999$. The gale fallback never triggers for these storms.

METRO MANILA BENCHMARK

Cutting every NCR road in NOAH 25-yr medium or high hazard leaves **29.7%** of NCR without a route within the cut-off rule. Tariverdi et al. (2023) report about 22% losing access to higher-level care at a 15 cm flood. The definitions differ: an order-of-magnitude check.

LEPTOSPIROSIS: THE STRETCH GOAL

The proposal assumed no open subnational leptospirosis series. Project CCHAIN (HDX) has weekly PIDSR counts for 12 cities, and the Mendeley (Parreño) file has national weeks. Both are analysed as **retrospective associations**, not forecasts: negative-binomial models of weekly cases on CHIRPS rain 0-6 weeks earlier, with seasonal terms, year and city effects and a COVID-19 term.

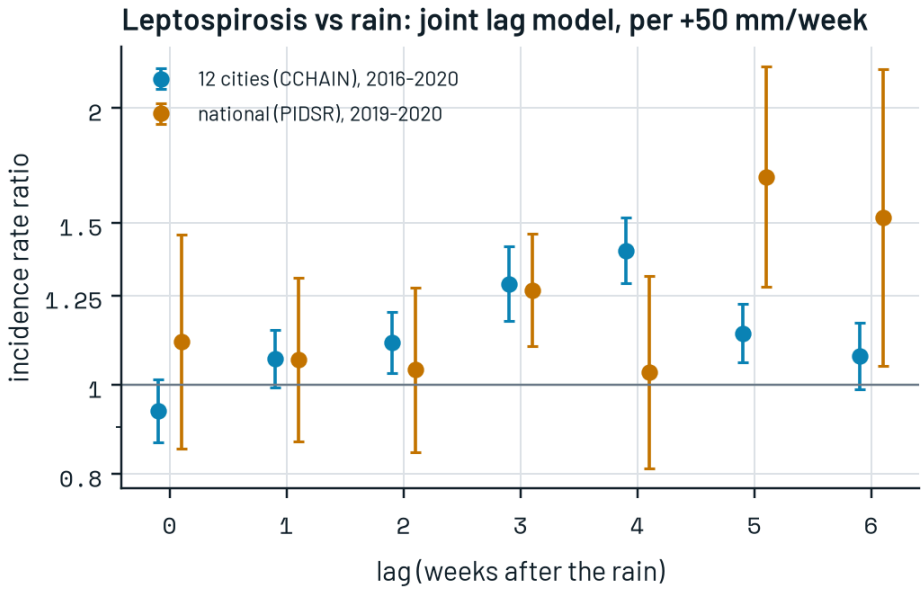


Figure 15. Joint distributed-lag incidence rate ratios (95% CI) per +50 mm/week of rain.

- **Cities (3,036 city-weeks, 1,685 cases, 2016-2020): the headline.** Rain is followed by leptospirosis 2-5 weeks later; the joint peak is lag 4, IRR 1.40 (1.29-1.52) per +50 mm/week, cumulative over lags 0-6 2.45 (2.05-2.92). A heavy-rain week (at or above the city's 90th percentile) peaks at lag 4: 2.00 (1.61-2.48). That is a week or two later than the ~2-week peak of Matsushita et al. (2018); the CCHAIN week label is uncertain by about a week.
- **National (80 weeks, 2019-2020): consistent, weaker.** Joint peak at lag 5: 1.68 (1.28-2.22); cumulative 4.12 (2.41-7.03). With 2019 alone the cumulative effect is 1.51 (0.76-3.00): much of the national signal comes from 2020 (the December surge after Rolly and Ulysses, under COVID reporting).
- **Typhoons: mixed.** After Rolly and Ulysses (2020), cities inside the gale swath had 4.31 (1.72-10.85) times the outside cities' observed / expected cases in weeks 1-4 (pre-landfall placebo 1.20, 0.32-4.53). For Ompong and Tisoy the ratio is 0.81 (0.55-1.20) but the placebo fails (2.36, 1.55-3.61): weeks of monsoon rain had already raised cases before Ompong. Four storms and 12 cities support no general storm effect.
- **Flood: not estimable.** Of 48 city-storm pairs, 31 had radar over more than 20% of the city and 8 show any new flood (at most 2.2%); too few to relate flood to cases. Flood mediation stays a literature assumption.

The 12 cities are not a sample of provinces, so nothing here is extrapolated to provinces, and leptospirosis is not in the dashboard's ranking.

PROPOSAL OBJECTIVES: SCORECARD

OBJECTIVE (PROPOSAL)	RESULT IN THIS BUILD	STATUS	NOTE
Province-week panel 2016-2020 on PSGC: dengue, rain, Sentinel-1, hazard, storms	85 harmonised units x 262 weeks; every listed layer plus river routing	done	85 harmonised 2016-2020 units, not 81 provinces (126 reporting units); Sentinel-1 only in the 4 storm windows
Lead-lag graph (PCMCI+), sentinels (PageRank), communities (Louvain); test sentinel features out of sample	8 links at q < 0.05; sentinels and communities on the dashboard; ablation run	done	tested and null: graph / no-graph relative WIS 0.999
Relative WIS below 1.0 vs seasonal naive, 2018-2019	XGBoost 0.544; median ensemble 0.444	done	
90% interval coverage of 80-95%	0.852 overall; 0.82-0.89 by horizon	done	4-week horizon close to the floor
Benchmarks: Chronos-2, TabPFN-TS, MPNN-LSTM; median ensemble; conformal check	scored on the same rows: Chronos-2, TabPFN-TS, MPNN-LSTM, median ensemble; online conformal check	done	the ensemble is evaluated, but storm scenarios use XGBoost (the zero-shot models take no storm-track input)
Access loss per municipality, friction-surface and network methods, four typhoons	both methods, 1,641 municipalities, 4 storms x 15 tracks	done	method disagreement reported as the A_lo-A_hi range
Sensitivity bounds (Kendall's τ); Metro Manila benchmark	9 variants x 2 methods, $\tau \geq 0.99$; NCR 30% vs ~22%	done	order-of-magnitude check
Dashboard: ranked provinces, calibrated risk, access, sentinels, drivers, tier, any historical or shifted track	60 backtest scenarios + 38 catalogue storms, pre-computed; R recalibrated	done	tiers are provisional rules
Backtest: did high-priority provinces see more cases?	ρ(R) 0.34-0.73 over 4 storms	done	2020 storms indicative only (COVID reporting)
System Usability Scale from three target users	test form and task built into the dashboard; no responses yet	not done	needs sessions with DOH / LGU / DRRM users
Leptospirosis (stretch): national lag analysis; region-week series if digitised	national lag analysis + an open 12-city series (CCHAIN) + typhoon event study	partly	association only, not in the forecast; no province-level series

PROPOSAL CORRECTIONS FOUND IN THE DATA

- 126 dengue reporting units, not 81 provinces (harmonised to 85); deaths unreliable; 2020 collapsed under COVID-19.
- An open subnational leptospirosis series exists (Project CCHAIN, 12 cities).
- EMSR478 (Rolly) has damage grading only, no flood delineation; the Global Flood Database (Tellman et al., 2021) has no Philippine event after 2017.
- Flood hazard maps come from the UP NOAH archive at 5 and 25 years, not LiPAD; 11 units have no 25-yr map.
- HydroRIVERS (Lehner & Grill, 2013) places the Philippines in the Australasia file, not Asia.
- roroph routes describe 2024-2026 operations with partly coarse port coordinates; NHFR blocks scripted download (OSM + NOAH fallback used).
- Meta HRSL under-counts people aged 60+ against the 2020 census.
- The proposal's final forecast was the median ensemble; storm scenarios use XGBoost (the only storm-conditioned model).

LIMITATIONS AND NEXT STEPS

THREATS TO VALIDITY

- **Two evaluation years.** 2018-2019 contains one epidemic; the model ranking changes between mean and median WIS because of it.
- **Surveillance artefacts.** Reporting delays and under-reporting are in the target; forecasts predict reported, not true, cases.
- **Observed flood.** Sentinel-1 sees each place every 6-12 days and misses flooding under dense vegetation and in built-up areas; shifted tracks and the catalogue fall back to static hazard maps.
- **Network gaps.** OSM completeness is uneven outside Luzon (the raster method bounds this); ferries are 2024-2026 routes; hospitals are an OSM / NOAH fallback without facility level or capacity. HRSL under-counts older people.
- **Provisional rules.** Tier thresholds, road speeds and the cut-off rule are assumptions, partly addressed by the sensitivity analysis.
- **Causality.** PCMCI+ links and the leptospirosis lags are associations, not transmission routes or causal effects.

NEXT STEPS

1. Run the System Usability Scale test with three DOH / LGU / DRRM users (Brooke, 1996) and review the tiers with them.
2. Feed a PAGASA forecast track and its cone in place of historical shifts.
3. Add a second radar or optical source between Sentinel-1 passes.
4. Replace the hospital fallback with the DOH facility registry (facility level, so access can mean a Level 2-3 hospital).
5. Give the benchmarks storm inputs, or blend the ensemble with XGBoost's storm signal, so scenarios can use the ensemble.

REPRODUCIBILITY

`setup.bat` (or `download_data.bat`) → `build.bat` → `run_app.bat`. This report and the slides are regenerated with `python scripts/make_results.py`. Missing inputs are skipped, listed in `data/processed/status.json` and on the dashboard; re-running after they arrive needs no code change.

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